

and more capable over time.

The rise of SDVs is made possible by a profound transformation in vehicle architecture. Traditional cars relied on a fragmented system of dozens of electronic control units (ECUs), each responsible for a specific function and connected through complex wiring networks. This approach, while effective in the past, created rigid systems in which innovation was slow and integration cumbersome. Modern vehicles are moving towards centralised computing architectures built around high-performance processors and zonal designs. Instead of dozens of isolated controllers, a few powerful computing units now orchestrate the entire vehicle. This shift simplifies hardware complexity while enabling unprecedented software flexibility. Companies like Tesla and Rivian have embraced this approach, designing vehicles more like distributed computing systems than mechanical assemblies. The result is a platform where software can be updated rapidly, features can scale seamlessly, and innovation cycles are dramatically accelerated.

Under the hood: A new technology stack

Beneath the surface, the modern SDV resembles a layered technology ecosystem rather than a traditional machine. “At its foundation lies advanced semiconductor hardware, providing the computational power required to process vast amounts of data in real time. This is complemented by power electronics that optimise energy usage, particularly in electric vehicles,” says Akshay Raj Mathur, Lead Product Manager at Stellantis.

Above this hardware layer sits a sophisticated software stack, including middleware and operating systems that abstract hardware complexities and enable application development. On top of this foundation, application layers deliver everything from infotainment

When software turns cars into living products

A SDV is not just a concept—it is already reshaping how cars behave, generate revenue, and evolve in the real world.

- **Performance upgrade without touching hardware.** A customer purchases a car with standard acceleration, but months later a software update unlocks faster acceleration without any mechanical changes. This enables the same hardware to deliver higher performance, allows manufacturers to monetize features after the initial sale, and extends the vehicle’s lifecycle value.
- **‘Dog mode’ and cabin intelligence.** A parked vehicle uses software to maintain a safe cabin temperature, monitor internal conditions, and display safety messages. This enhances safety without additional hardware, builds emotional trust with users, and turns small software-driven features into meaningful brand differentiators.
- **Over-the-air (OTA).** Vehicles now receive over-the-air updates that can fix bugs, improve battery efficiency, or even add entirely new features overnight. Modern cars already run over 100 million lines of code, with future autonomous systems expected to exceed 500 million. This shift eliminates dependence on dealerships, enables continuous performance improvement, and significantly reduces recall costs.
- **AI-driven personalisation.** The vehicle learns driver habits such as preferred temperature settings, driving style, and frequent routes, and automatically adapts over time. This creates a personalised experience without manual input, improves user satisfaction and retention, and enables continuous, data-driven feature evolution.
- **Predictive and connected ecosystems.** Fleet vehicles leverage geofencing, real-time alerts, and predictive maintenance to optimise operations, reducing downtime, lowering costs, and improving safety and efficiency. Unlike traditional cars that depreciate immediately after purchase, SDVs can improve over time, gain new capabilities, and generate continuous revenue.

It is no longer just a product. It is a platform that evolves.

experiences to advanced driver-assistance systems. Cloud connectivity extends the vehicle’s capabilities beyond its physical boundaries, enabling continuous updates, remote diagnostics, and data-driven services.

At the highest level, artificial intelligence introduces adaptability and learning, transforming the vehicle into a system that can interpret, predict, and respond intelligently to its environment. Companies such as Nvidia and Qualcomm are now central players in this ecosystem, underscoring how deeply technology has become intertwined with automotive innovation.

To better understand this transformation, it is useful to visualise a simplified SDV architecture. At the lowest layer sits the hardware foundation, comprising high-performance SoCs, sensors, actuators, and power electronics. Above this lies the communication backbone, where traditional CAN buses are

increasingly complemented or replaced by high-speed automotive Ethernet, enabling faster data exchange across zonal architectures.

On top of this hardware layer, middleware operates, abstracting hardware complexity and enabling interoperability between different software components. Platforms such as AUTOSAR (Automotive Open System Architecture), both Classic and Adaptive, play a critical role here by standardising interfaces and enabling scalable software development across suppliers.

Above the middleware sits the operating system and application layer, where features such as infotainment, ADAS, and vehicle control logic reside. Cloud connectivity extends this stack beyond the vehicle, enabling OTA updates, diagnostics, and data-driven services. This layered approach transforms the vehicle into a modular, upgradable computing platform rather than a fixed-function machine.